AI-ASSISTED CODING
2303A510A0 - VISHNU VARDHAN
Assignment - 17.1
Batch - 14

Lab 17 – AI for Data Processing: Data Cleaning and Preprocessing Scripts

Task 1 – Customer Data Cleaning

Prompt:

Generate a Python script to clean a customer dataset. Handle missing values in age, income, city columns, remove duplicates, standardize city names to lowercase, and encode categorical variables gender and city.

Code:

```
import pandas as pd
from sklearn.preprocessing import LabelEncoder
data = pd.read_csv("/content/shopping_trends.csv")
data['Age'].fillna(data['Age'].median(), inplace=True)
data.drop_duplicates(inplace=True)
data['Location'] = data['Location'].str.lower()
le = LabelEncoder()
data['Gender'] = le.fit_transform(data['Gender'])
data['Location'] = le.fit_transform(data['Location'])
print(data.head())
```

Output:

```
   Customer ID  Age  Gender Item Purchased  Category  Purchase Amount (USD)  \
0            1   55       1         Blouse  Clothing                     53
1            2   19       1        Sweater  Clothing                     64
2            3   50       1          Jeans  Clothing                     73
3            4   21       1        Sandals  Footwear                     90
4            5   45       1         Blouse  Clothing                     49

   Location Size      Color  Season  Review Rating Subscription Status  \
0        16    L       Gray  Winter            3.1                 Yes
1        18    L     Maroon  Winter            3.1                 Yes
2        20    S     Maroon  Spring            3.1                 Yes
3        38    M     Maroon  Spring            3.5                 Yes
4        36    M  Turquoise  Spring            2.7                 Yes

  Payment Method  Shipping Type Discount Applied Promo Code Used  \
0    Credit Card        Express              Yes             Yes
1  Bank Transfer        Express              Yes             Yes
2           Cash  Free Shipping              Yes             Yes
3         PayPal   Next Day Air              Yes             Yes
4           Cash  Free Shipping              Yes             Yes

   Previous Purchases Preferred Payment Method Frequency of Purchases
0                  14                    Venmo            Fortnightly
1                   2                     Cash            Fortnightly
2                  23              Credit Card                 Weekly
3                  49                   PayPal                 Weekly
4                  31                   PayPal               Annually
```

Explanation:

- Missing values replaced using median/mode
- Duplicate rows removed
- City names converted to lowercase
- Gender and city converted to numeric form for ML models

Task 2 – Sales Data Preprocessing

Prompt:

Generate a Python script to preprocess sales data.
Convert date column to datetime, create month/year/day features, normalize sales amount, and remove outliers.

Code:

```
import pandas as pd
from sklearn.preprocessing import MinMaxScaler
data = pd.read_csv("/content/sales_data.csv")
data.columns = data.columns.str.strip()
print("Columns in dataset:")
print(data.columns)
date_col = None
for col in data.columns:
    if "date" in col.lower():
        date_col = col
        break
if date_col is None:
    print("No date column found in dataset")
else:
    print("Date column found:", date_col)
    data[date_col] = pd.to_datetime(data[date_col])
    data['Month'] = data[date_col].dt.month
    data['Year'] = data[date_col].dt.year
    data['DayOfWeek'] = data[date_col].dt.dayofweek
sales_col = None
for col in data.columns:
    if "sale" in col.lower() or "revenue" in col.lower():
        sales_col = col
        break
if sales_col is not None:
    scaler = MinMaxScaler()
    data[sales_col] = scaler.fit_transform(data[[sales_col]])
    print("Normalized:", sales_col)
else:
    print("No sales column found")
print(data.head())
```

Output:

```
   Product_ID  Sale_Date Sales_Rep Region  Sales_Amount  Quantity_Sold  \
0        1052   0.090411       Bob  North       5053.97             18
1        1093   0.301370       Bob   West       4384.02             17
2        1015   0.720548     David  South       4631.23             30
3        1072   0.643836       Bob  South       2167.94             39
4        1061   0.224658   Charlie   East       3750.20             13

  Product_Category  Unit_Cost  Unit_Price Customer_Type  Discount  \
0        Furniture     152.75      267.22     Returning      0.09
1        Furniture    3816.39     4209.44     Returning      0.11
2             Food     261.56      371.40     Returning      0.20
3         Clothing    4330.03     4467.75           New      0.02
4      Electronics     637.37      692.71           New      0.08

  Payment_Method Sales_Channel Region_and_Sales_Rep  Month  Year  DayOfWeek
0           Cash        Online            North-Bob      2  2023          4
1           Cash        Retail             West-Bob      4  2023          4
2  Bank Transfer        Retail          South-David      9  2023          3
3    Credit Card        Retail            South-Bob      8  2023          3
4    Credit Card        Online         East-Charlie      3  2023          4
```

Explanation:

- Datetime conversion enables time analysis
- Feature engineering improves model performance
- Normalization scales data uniformly
- IQR method removes extreme values

Task 3 – Healthcare Data Preparation

Prompt:

Generate a Python script to preprocess healthcare data by handling missing values, encoding categorical features, scaling numerical data, and splitting into training and testing sets.

Code:

```
import pandas as pd
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.model_selection import train_test_split
data = pd.read_csv("/content/dirty_v3_path.csv")
data.columns = data.columns.str.strip()
print("Columns in dataset:")
print(data.columns)
num_cols = data.select_dtypes(include=['int64', 'float64']).columns
for col in num_cols:
    data[col].fillna(data[col].mean(), inplace=True)
cat_cols = data.select_dtypes(include='object').columns
for col in cat_cols:
    data[col].fillna(data[col].mode()[0], inplace=True)
le = LabelEncoder()
for col in cat_cols:
    data[col] = le.fit_transform(data[col])
scaler = StandardScaler()
data[num_cols] = scaler.fit_transform(data[num_cols])
X = data.iloc[:, :-1]
y = data.iloc[:, -1]
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)
print("Training shape:", X_train.shape)
```

Output:

```
Columns in dataset:
Index(['Age', 'Gender', 'Medical Condition', 'Glucose', 'Blood Pressure',
       'BMI', 'Oxygen Saturation', 'LengthOfStay', 'Cholesterol',
       'Triglycerides', 'HbA1c', 'Smoking', 'Alcohol', 'Physical Activity',
       'Diet Score', 'Family History', 'Stress Level', 'Sleep Hours',
       'random_notes', 'noise_col'],
      dtype='object')
/tmp/ipykernel_17029/1863719885.py:10: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inp
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead,


  data[col].fillna(data[col].mean(), inplace=True)
/tmp/ipykernel_17029/1863719885.py:13: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inp
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead,


  data[col].fillna(data[col].mode()[0], inplace=True)
Training shape: (22340, 19)
```

Explanation:

- Missing medical values replaced using mean
- Categorical variables converted to numeric
- Standard scaling improves ML accuracy
- Train-test split prevents overfitting

TASK 4 – Real-Time Application: Multi-Language Snippet Converter

Prompt:

Generate a Python script to preprocess employee salary data by handling missing values, removing salary outliers, standardizing department names, and encoding location and department.

Code:

```
import pandas as pd
from sklearn.preprocessing import LabelEncoder
data = pd.read_csv("/content/Employee_Salary.csv")
data.fillna(method='ffill', inplace=True)
Q1 = data['Salary'].quantile(0.25)
Q3 = data['Salary'].quantile(0.75)
IQR = Q3 - Q1
data = data[(data['Salary'] >= Q1 - 1.5 * IQR) &
            (data['Salary'] <= Q3 + 1.5 * IQR)]
data['Department'] = data['Department'].str.lower()
le = LabelEncoder()
data['Location'] = le.fit_transform(data['Location'])
data['Department'] = le.fit_transform(data['Department'])
print(data.head())
```

Output:

	Employee_ID	Department	Job_Title	Experience	Education	Location	Salary
0	101	1	Manager	10	MBA	2	75000
1	102	2	Developer	5	B.Tech	4	60000
2	103	0	Analyst	7	MBA	1	65000
3	104	2	Developer	3	B.Tech	0	55000
4	105	1	Executive	4	BBA	2	50000

Explanation:

- Missing values filled using forward fill
- IQR removes unrealistic salary values
- Lowercase standardizes department names
- Encoding converts text to numbers for analytics